

Separation Anxiety

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Logistic regression

Models a binary outcome variable y based on predictor/explanatory variables $\{x_i\}$

Outcome $y_i \in \{0,1\}$

$$y_i \sim \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i}$$

Distribution: $y_i \sim \text{Bernoulli}(p_i)$

Logistic regression is applied within fields such as political science, medicine, econometrics, machine learning (binary classification), and more.

Logit Link

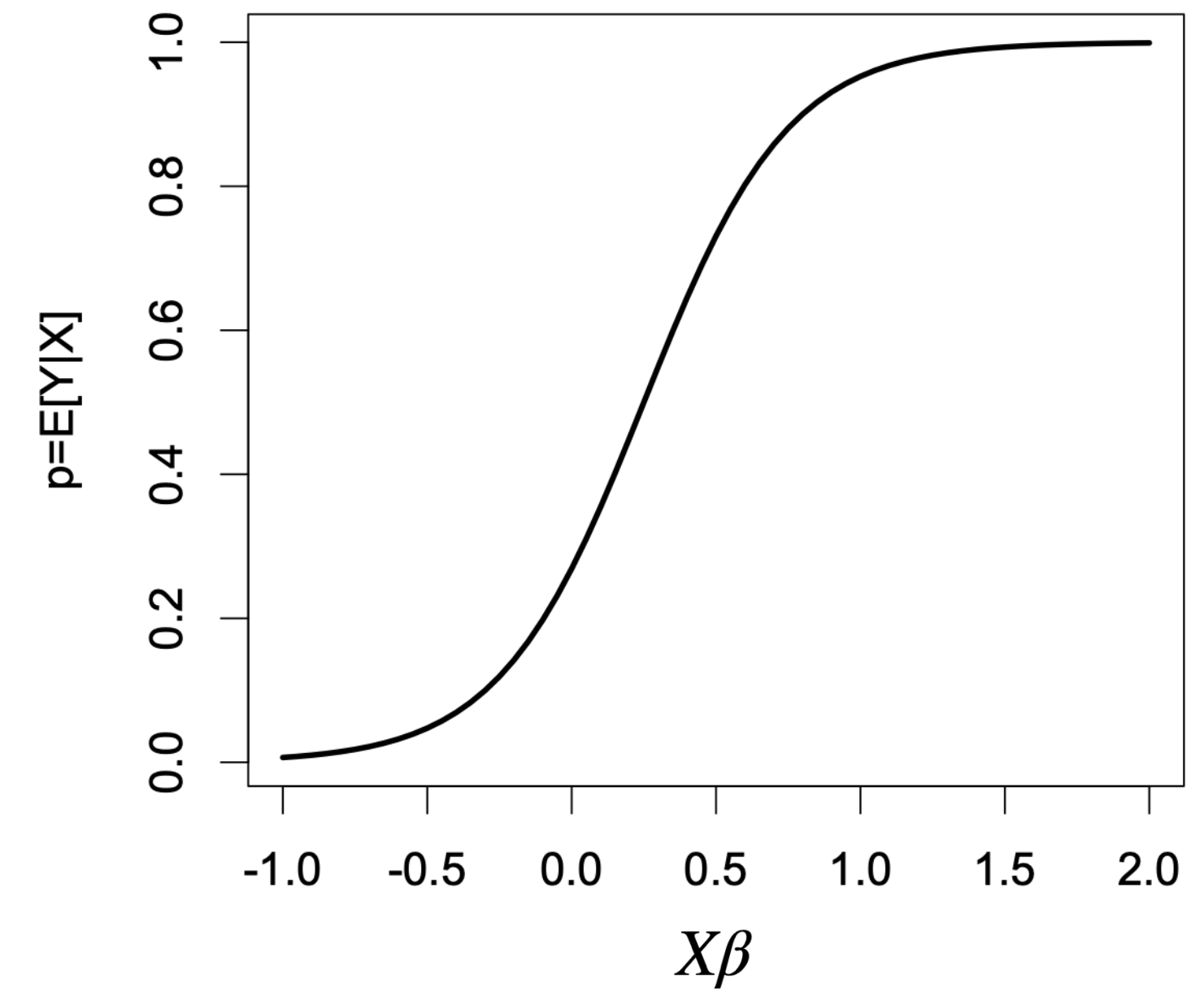
$$\Pr(y_i = 1) = p_i = \frac{1}{1 + e^{-X_i\beta}} = \frac{e^{X_i\beta}}{1 + e^{X_i\beta}}$$

$$\Pr(y_i = 0) = 1 - p_i = \frac{e^{-X_i\beta}}{1 + e^{-X_i\beta}} = \frac{1}{1 + e^{X_i\beta}}$$

$$\ln L = \sum [y_i \ln p_i + (1 - y_i) \ln(1 - p_i)]$$

$$= \sum \left[y_i \ln \left(\frac{1}{1 + e^{-X_i\beta}} \right) + (1 - y_i) \ln \left(\frac{1}{1 + e^{X_i\beta}} \right) \right]$$

$$= \sum [-y_i \ln(1 + e^{-X_i\beta}) - (1 - y_i) \ln(1 + e^{X_i\beta})]$$



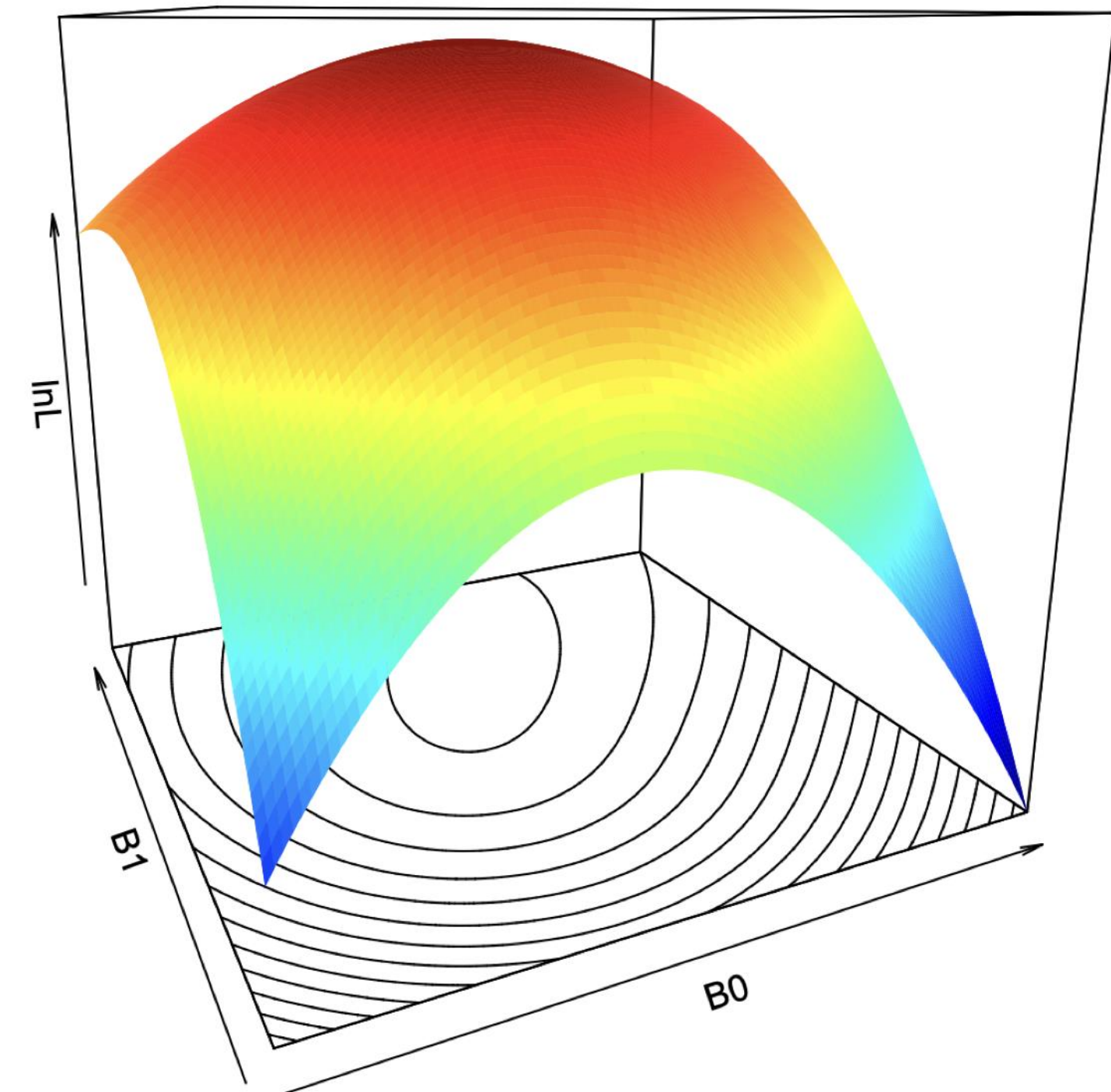
Logit Log-Likelihood

$$\begin{aligned}\ln L &= \sum \left[-y_i \ln(1 + e^{-X_i\beta}) - (1 - y_i) \ln(1 + e^{X_i\beta}) \right] \\ &= \sum \left[-y_i \ln \left(1 + e^{-(\beta_0 + \beta_1 x_{1i})} \right) - (1 - y_i) \ln \left(1 + e^{(\beta_0 + \beta_1 x_{1i})} \right) \right]\end{aligned}$$

Hill-shaped function of the *parameters*.

$\hat{\beta}$'s found by maximizing the $\ln L$

Usually via iterated algorithms for hill-climbing or canned commands that do so — e.g., `glm()`



What is separation?

In logistic regression with dummy variable X:

- Separation $\neq$ totally perfect prediction of all Y observations
- Separation = perfect prediction of a particular Y outcome {0 or 1} by a particular X value {0 or 1}

Can be a problem for MLE for Binary and other data types

- Multinomial/categorical outcomes, counts, tobit, ...

What does separation look like?

$$y \sim \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

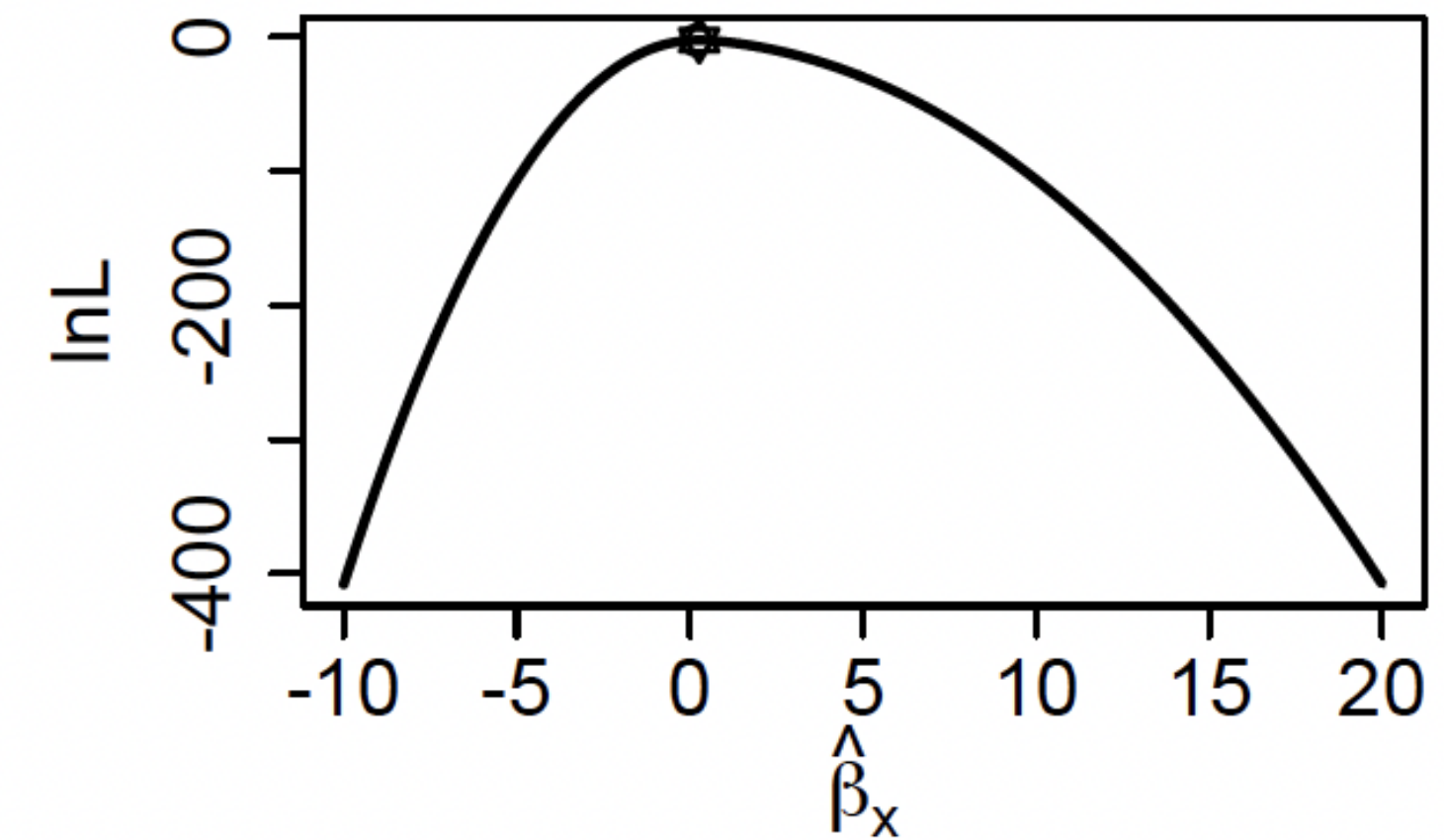
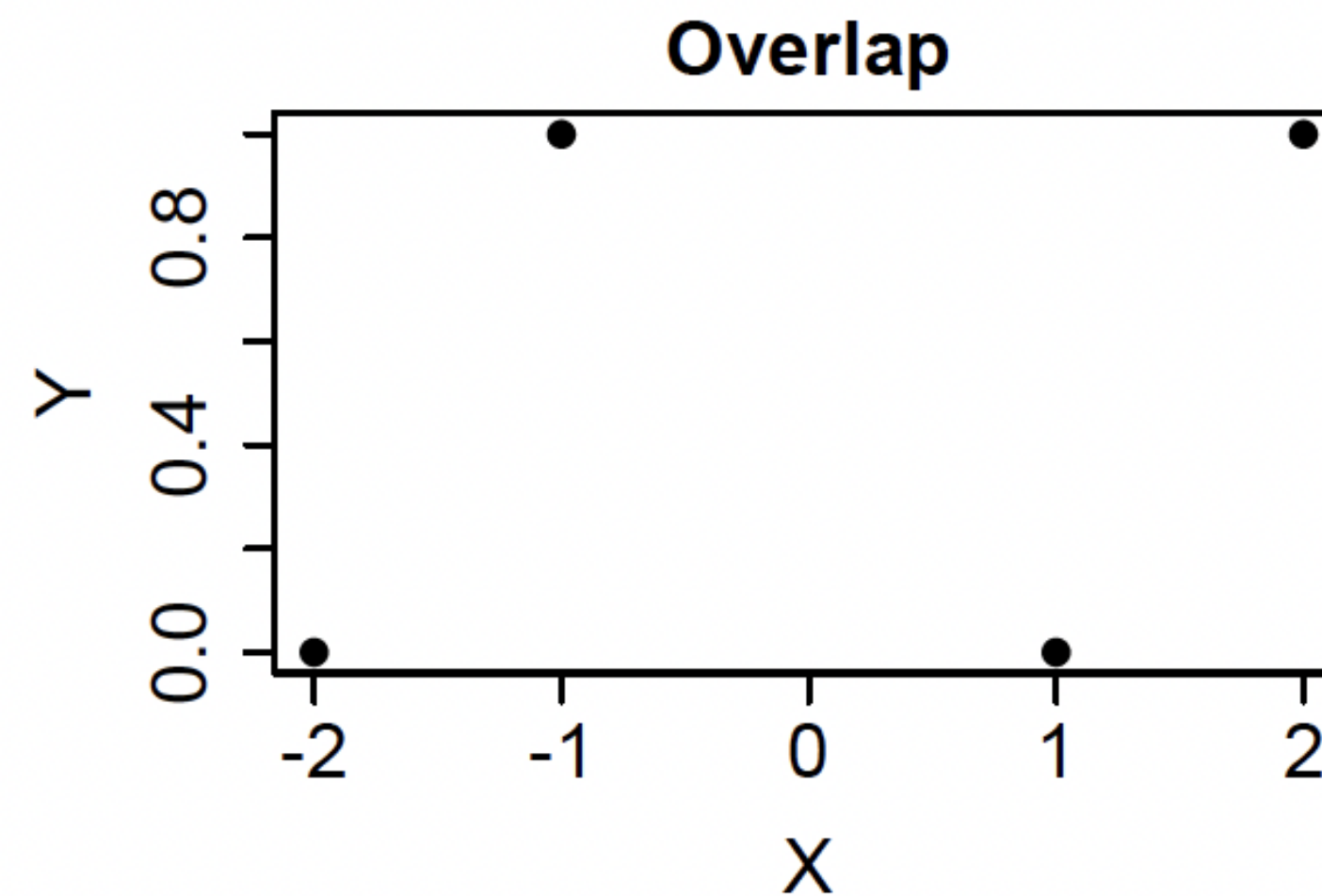
Observation i	y_i	x_1	x_2	x_3
1	1	1	0	0
2	0	1	0	0
3	1	0	1	0
4	0	1	0	0
5	0	0	1	0
6	1	0	0	1

Is there a value of x_1 that perfectly predicts a value of y ?

x_2 ? x_3 ?

When separation does not occur

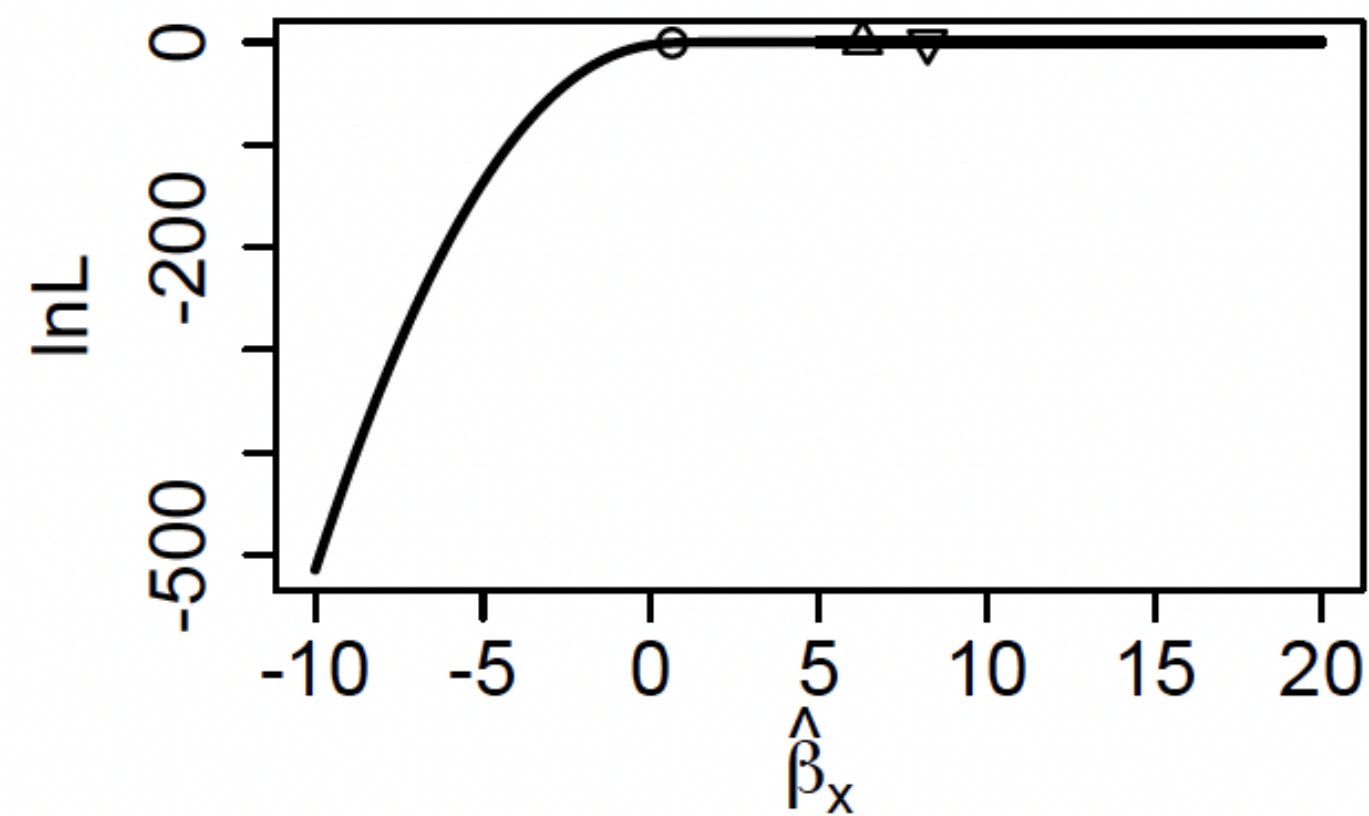
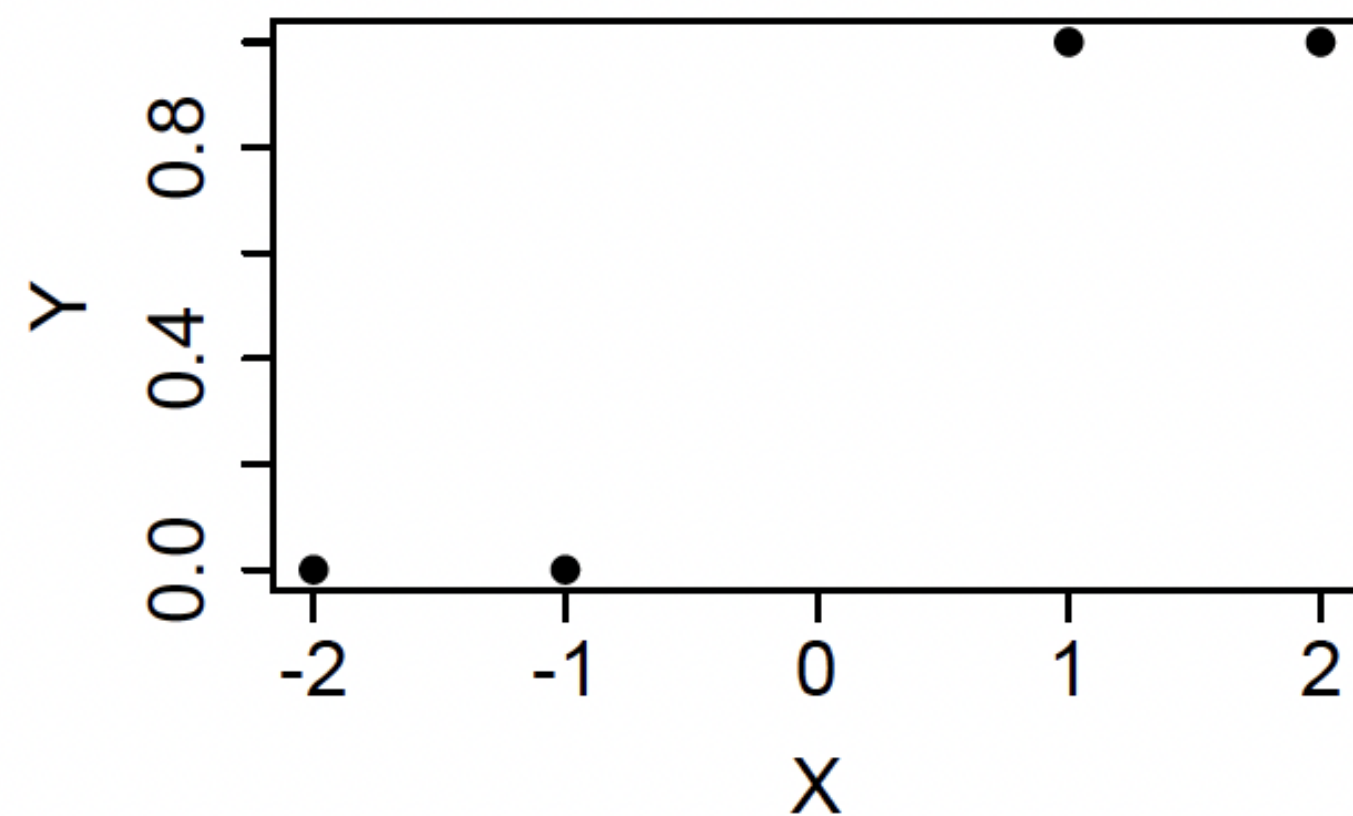
Y	X
0	-2
1	-1
0	1
1	2



1. $\ln L$ has a unique maximum at $\hat{\beta}_x$
2. We can estimate the std error of $\hat{\beta}_x$

When separation occurs

Perfect Separation



In both cases:

1. lnL doesn't have a unique maximum. It is monotonically

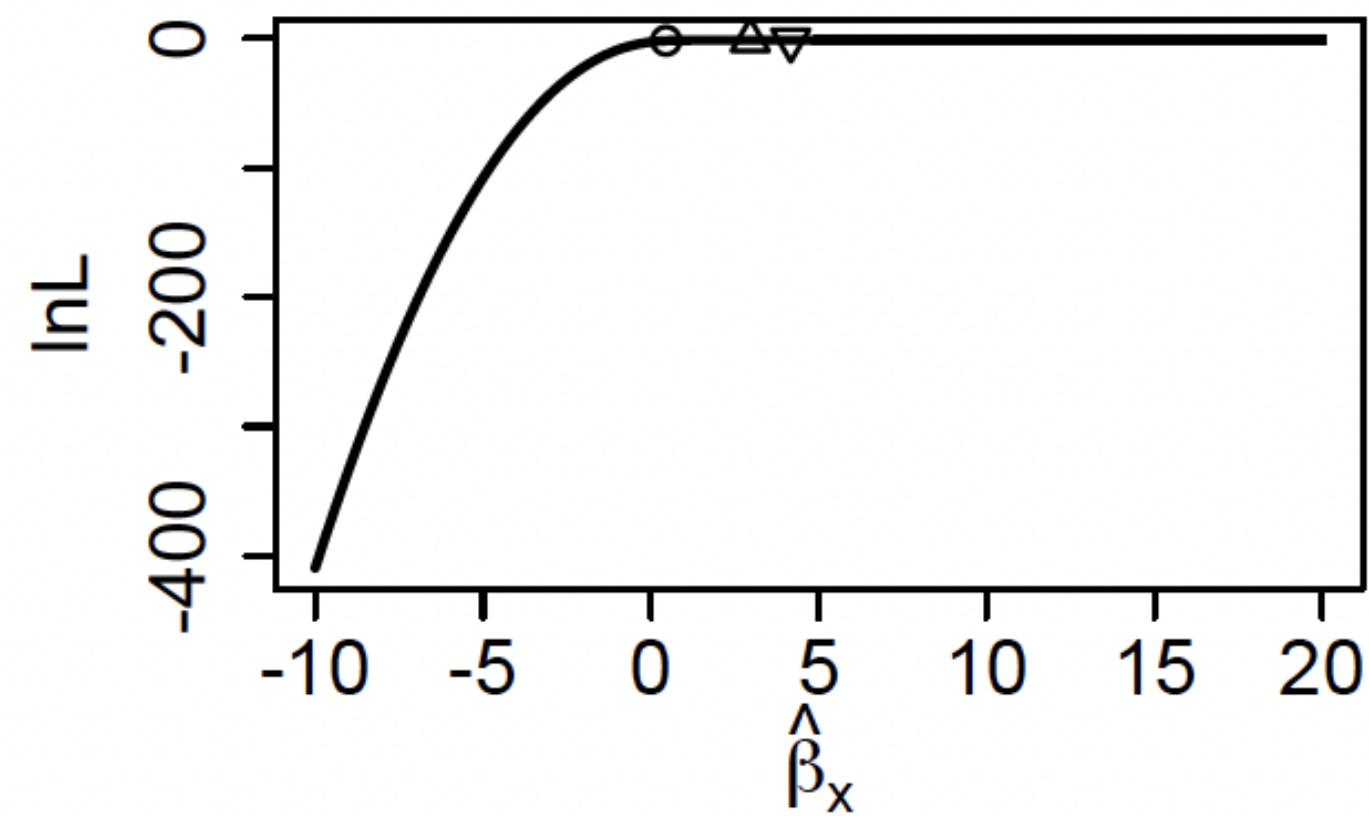
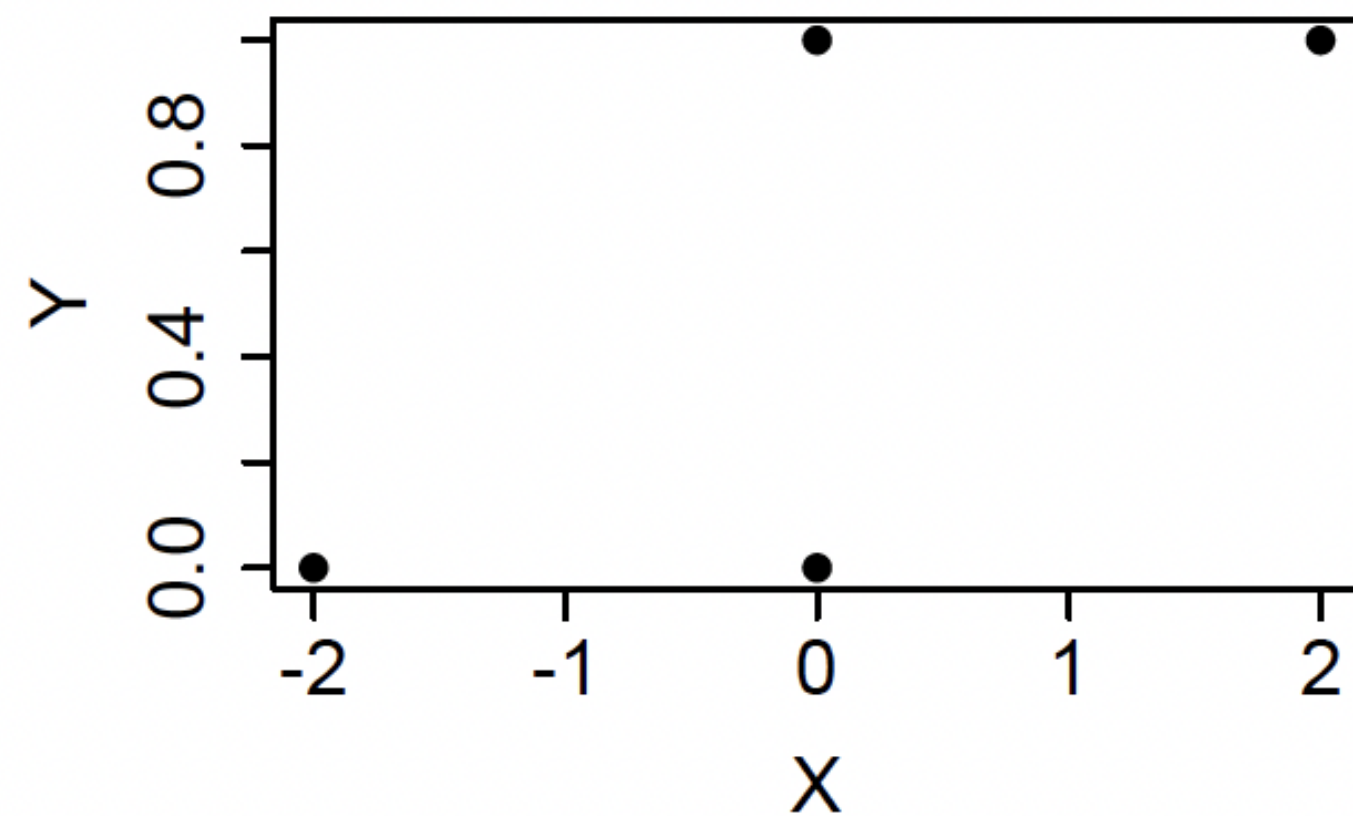
increasing in $\hat{\beta}_x$ to an asymptote.



2. We can't estimate the std error of $\hat{\beta}_x$



Quasi-Complete Separation



Canned statistical commands

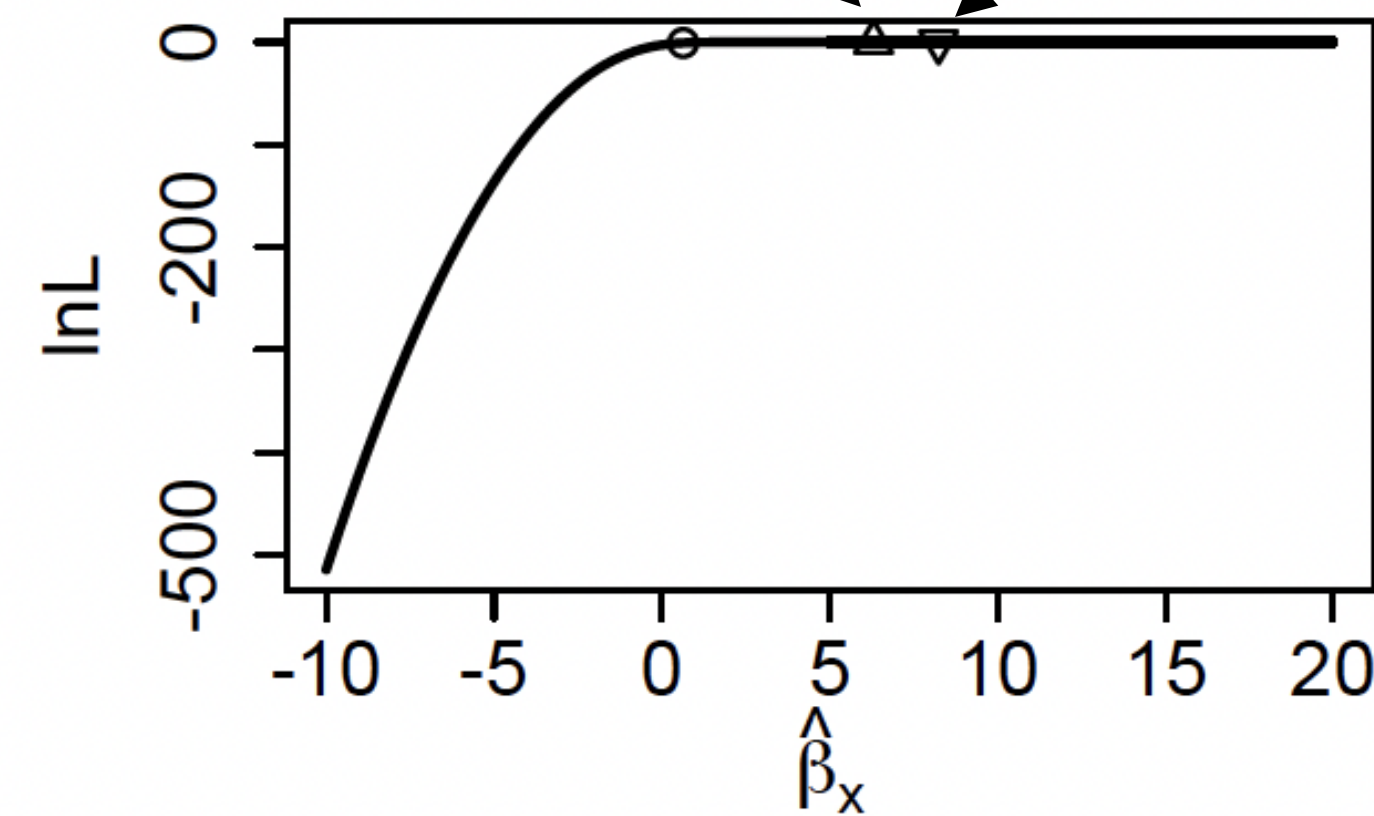
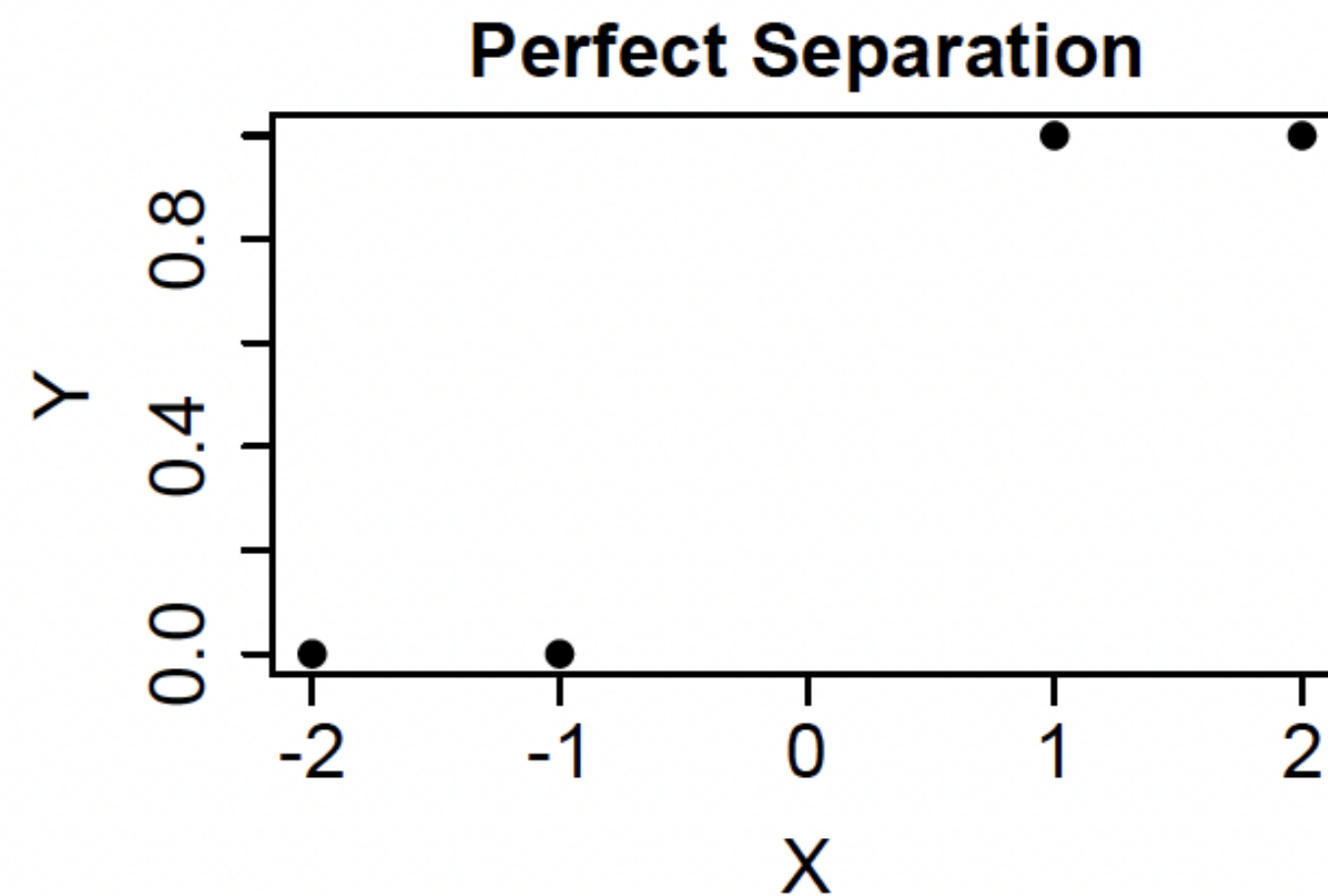
Unfortunately, canned statistical commands may not catch the separation

- May report results without providing any warning
- Researchers can detect separation by observing incorrectly large coefficient estimates, large standard errors, and large Wald confidence intervals

How does MLE fail with separation?

Using default
convergence
threshold produces
an estimate here.

Making the
convergence
threshold more
stringent moves the
estimate.

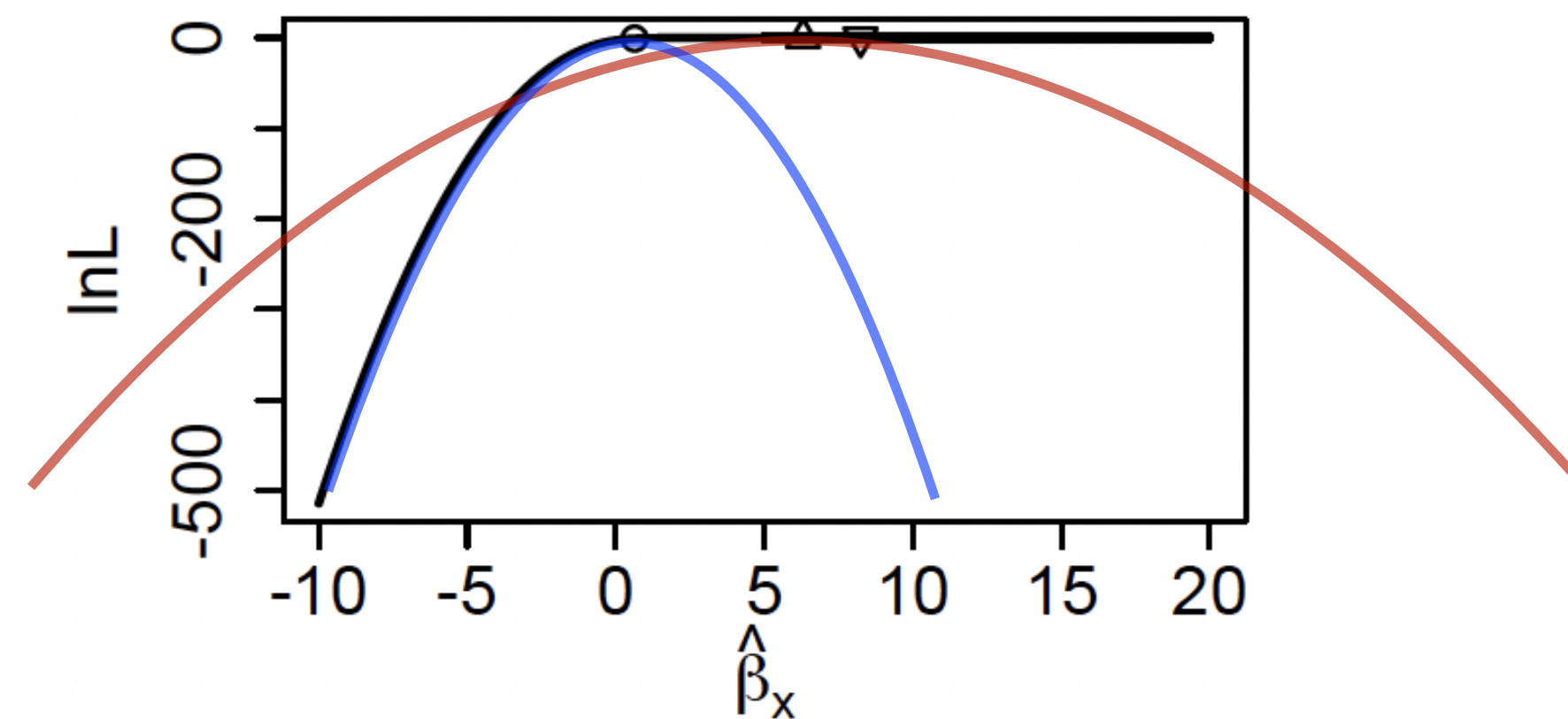
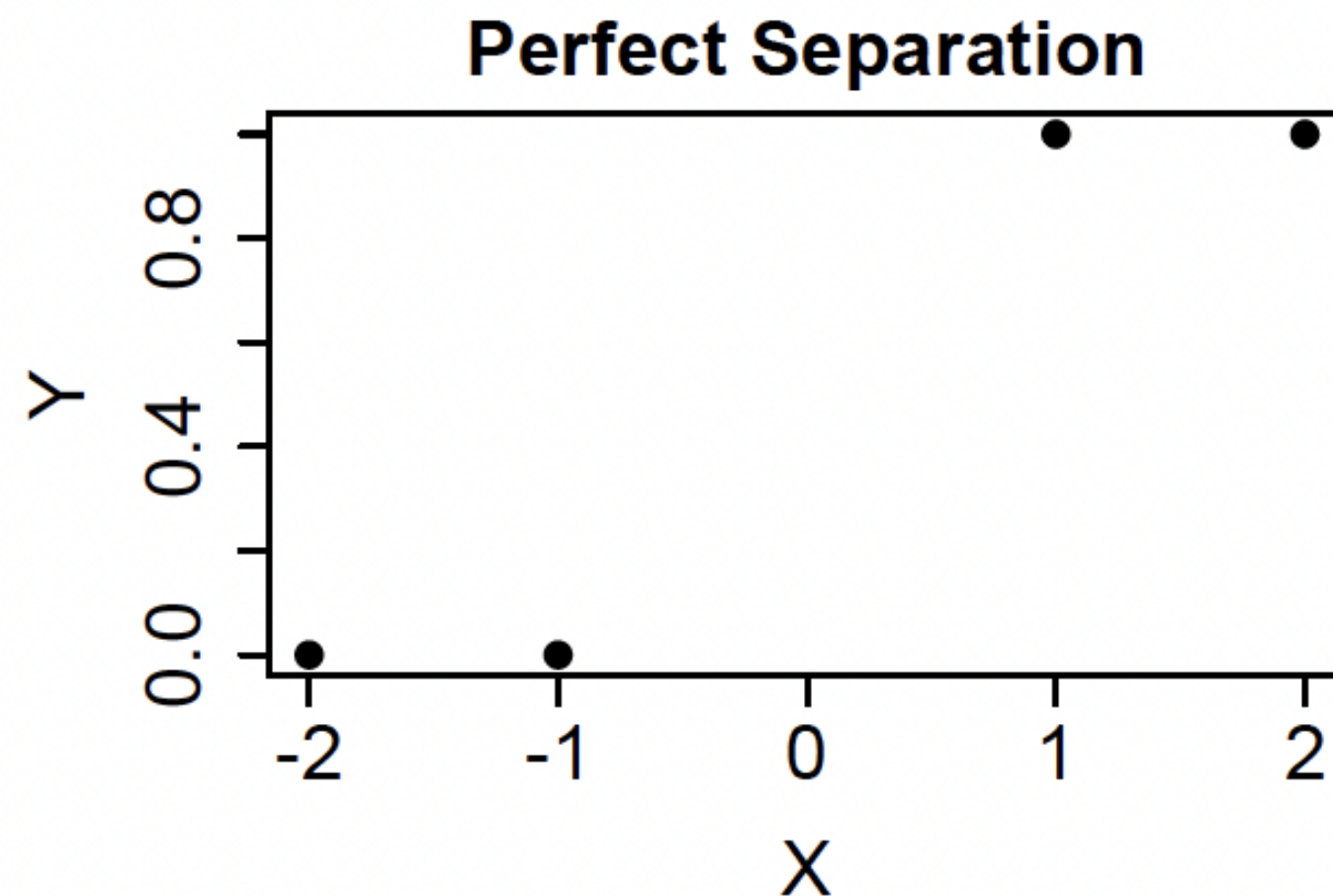


How does MLE fail with separation?

With MLE, we typically estimate std errors using a quadratic approximation to the $\ln L$.

With separation, the $\ln L$ is monotonically increasing in $\hat{\beta}_x$.

The resulting std error will often appear to be huge (red) compared to no separation (blue).



What can lead to separation?

In the presence of a rare outcome or rare exposures

When there are highly correlated covariates

When there are covariates with strong effects

When the sample size is small

When there are more independent variables, especially if there are many dichotomous independent variables

Review so far...

In a logistic regression (binary Y variable) with a dummy X, separation occurs when a value of X is associated with only one value of Y.

When X is not a dummy, perfect (or complete) separation occurs when there is a threshold value X^* such that $X > X^*$ values are associated with only one value of Y and $X \leq X^*$ values are associated with the other value of Y. Quasi-complete separation replaces the strict equality $X > X^*$ with $X \geq X^*$.

When there is no separation, the lnL has a unique maximum and we can estimate std errors using the typical quadratic approximation to the lnL.

When separation occurs, the lnL increases monotonically to an asymptote for the estimate/variable inducing the separation.

- No MLE exists for that estimate. However, canned commands may still report one.
- No std error exists for that estimate. However, canned commands may still report one.
- Using the reported estimate and std error for inference may result in incorrect conclusions.

Poor solutions

Omitting the variable with separation

- Omitted variable bias: impact of omitted variable may be misattributed to other explanatory variables
- Doesn't make sense to get rid of data that explain the outcome well

Ignoring the separation

- Leads researchers to overstate the magnitude of the effects (by interpreting large coefficient estimates)

Switching to a different type of model

- Ex. Switching to OLS
- May not be as suitable for inference and prediction

Current solutions

Firth's bias reduced estimator

- Developed to reduce the small sample bias of MLE in logistic regression.
- Penalized $\ln L$: $\ln L^* = \ln L + (1/2)\ln|I(\beta)|$.

Bayesian priors

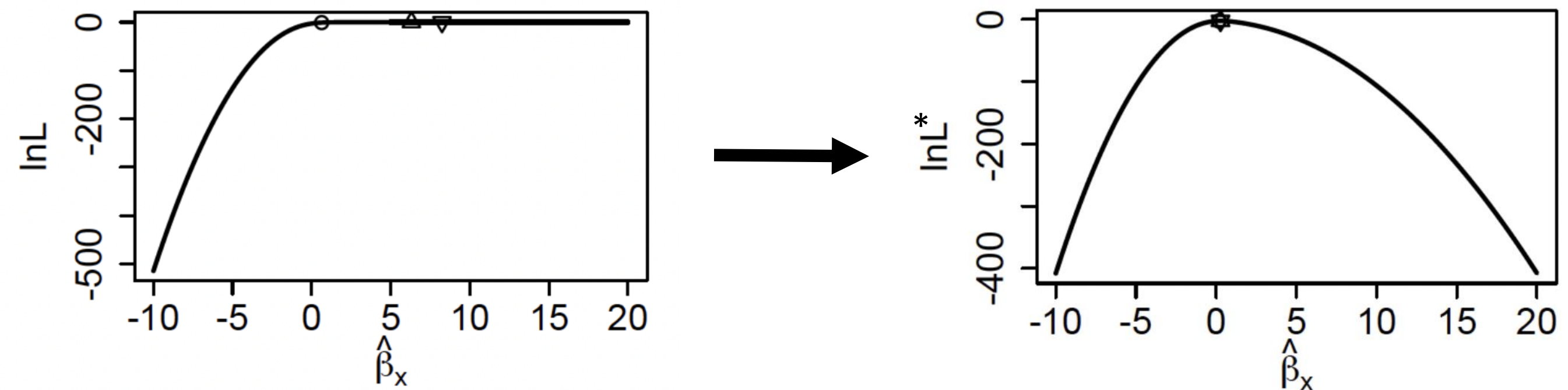
- Jeffrey's, Normal, Cauchy

Less investigated methods

- Ex. Hidden logistic regression models, exact logistic regression

If possible, increasing the sample size

- However, it has been shown that this may be less efficient than simply using Jeffreys' prior on the original sample



Current solutions

Profile likelihood based Confidence Interval

To address issues with the Wald CI, we compute a **profile likelihood** for β_1 :

$$\ell_p(\beta_1) = \max_{\beta_0} \sum_{i=1}^n [y_i \log (\Lambda(\beta_0 + \beta_1 x_i)) + (1 - y_i) \log (1 - \Lambda(\beta_0 + \beta_1 x_i))]$$

where $\Lambda(z) = \frac{1}{1+e^{-z}}$ is the logistic function.
We define the **likelihood ratio statistic**:

$$\lambda(\beta_1) = 2 \left[\ell_p(\hat{\beta}_1) - \ell_p(\beta_1) \right]$$

The **one-sided 95% lower confidence bound** for β_1 is given by:

$$\lambda(\beta_1) \leq \chi^2_{1,0.95}$$

This yields an interval:

$$\beta_1 \in [\beta_L, \infty), \quad \text{where } \beta_L = \inf \left\{ \beta_1 : \lambda(\beta_1) \leq \chi^2_{1,0.95} \right\}$$

Logit Regression Results						
=====						
Dep. Variable:	y	No. Observations:	100			
Model:	Logit	Df Residuals:	98			
Method:	MLE	Df Model:	1			
Date:	Wed, 06 Aug 2025	Pseudo R-squ.:	1.000			
Time:	15:20:39	Log-Likelihood:	-1.8764e-08			
converged:	False	LL-Null:	-69.315			
Covariance Type:	nonrobust	LLR p-value:	5.308e-32			
=====						
	coef	std err	z	P> z	[0.025	0.975]

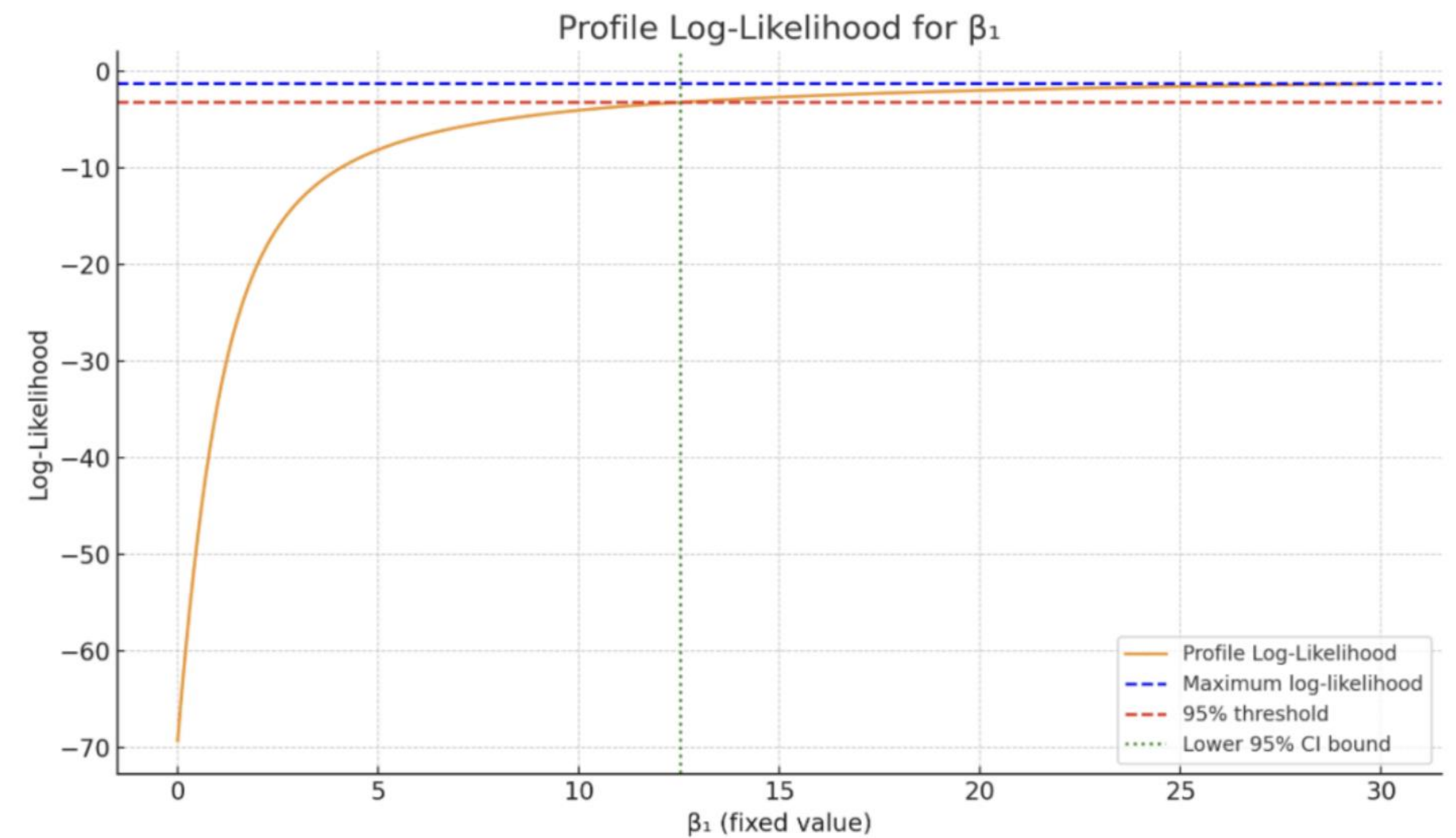
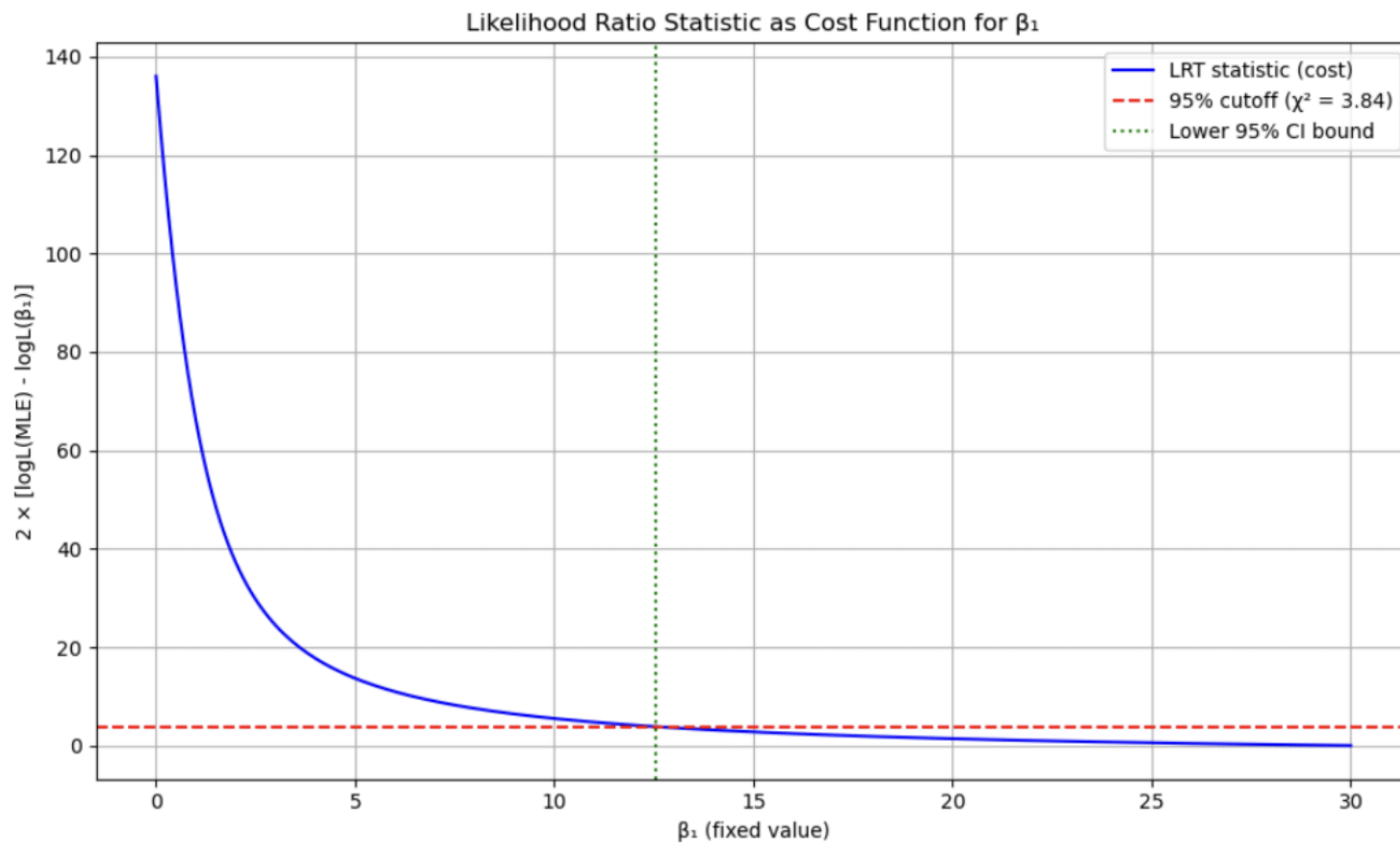
const	1.882e-09	7300.284	2.58e-13	1.000	-1.43e+04	1.43e+04
x1	914.9820	3.61e+05	0.003	0.998	-7.07e+05	7.09e+05
=====						
Complete Separation: The results show that there iscomplete separation or perfect prediction. In this case the Maximum Likelihood Estimator does not exist and the parameters are not identified.						

Wald Confidence Interval	
β_0	β_1
$[-14308.29, 14308.29]$	$[-707345.56, 709175.52]$

Table 1: Wald CI for intercept and slope under separation

Current solutions

Profile likelihood based Confidence Interval



Should we care about separation?

Separation is an issue for inference and estimation

- Separation is an issue when it comes to parameter estimates, confidence intervals, and drawing conclusions about the relationship between predictors and the outcome variable.

Separation may not be an issue for prediction

- When separation occurs, a model can still achieve high prediction accuracy, even if model coefficients are infinite.

MC analysis

- If we don't care about the regressor with separation, coefficients on other regressors will be unbiased, even if those regressors are correlated with the separating variable.
- Can conduct inference on the other variables in the model.

Our conclusion

Use a variety of techniques when facing separation

- See if the results agree with each other

Do inference based on profile penalized likelihood ratio tests

- When separation occurs, parameter estimate distributions can be very skewed
- Usually, Wald tests are used for inference (ex. For confidence intervals)
- Wald tests assume a symmetric distribution, so they fail when separation occurs

Future work

Exploring more with likelihood profiling

- Likelihood profiling has been used for other models with incomplete data
- Since typical Wald tests do not perform well in the presence of separation, profile likelihood methods for inference are typically used.

Adding noise to the model, then correcting for it

- It could also be a measurement error model that adds noise after the outcomes are initially calculated.
- Hope for both these cases is to induce overlap without destroying unbiasedness and efficiency

Experimenting more with data manipulation

- In current research, data manipulation is not seen as a viable solution to separation
- However, we think that some forms of data manipulation, such as using weights, could help solve separation without skewing the estimates too much.
- No one has settled on a way to tune weights so that Wald or profile-likelihood CIs keep their nominal coverage when N is tiny.

Thank you!